

Module Theory

Prelude

Exercise 1.

- (a) The element 2 is invertible in both $\mathbb{Z}[\frac{1}{2}]$ and in $\mathbb{Z}/3\mathbb{Z}$. Find a homomorphism $\phi: \mathbb{Z}[\frac{1}{2}] \rightarrow \mathbb{Z}/3\mathbb{Z}$ which sends $2^{-1} \mapsto 2^{-1}$.
- (b) Find an element in $\ker \phi$ which is not in $\mathbb{Z}$.

Exercise 2. Recall that if R is a ring and $S \subset R$ is a multiplicative subset, then in the localization $S^{-1}R$ two fractions $\frac{r_1}{s_1}, \frac{r_2}{s_2}$ are equal to one another if there is some $s_3 \in S$ so that $s_3(r_1s_2 - r_2s_1) = 0$. For the rest of this problem, let $R = \mathbb{Z}/12\mathbb{Z}$, and consider the localizations R_f for various $f \in R$.

- (a) For a given f , which fractions $\frac{n}{fk}$ are equal to $\frac{0}{1}$ in the localization?
- (b) Prove that for a given f , in R_f we have $\frac{1}{f} = \frac{y}{1}$ for some $y \in R$. Conclude that every $\frac{n}{fk}$ is equal to a fraction with 1 in the denominator. (Hint: rephrase this in terms of something being equal to 0.)
- (c) Which rings can arise as localizations R_f ? For each possible ring, which f s achieve that ring?
- (d) Draw the lattice of ideals in R , and the lattice of ideals in each possible R_f . What do you notice comparing them?

Exercise 3. (January 2020 Problem 3) Let R be a commutative ring, I an ideal in R , and S a multiplicative subset of R . There are two standard correspondences involving prime ideals:

- (i) Prime ideals in R/I are in bijection with prime ideals in R which contain I .
- (ii) Prime ideals in $S^{-1}R$ are in bijection with prime ideals in R which do not meet S .
- (a) Prove (i) and (ii). (The original problem said “or”, but in the SEP you have no choice.)
- (b) Give an example of a ring with finitely many elements that has exactly three prime ideals.
- (c) Give an example of a subring of $\mathbb{Q}$ that has exactly three prime ideals.

Exercise 4.

- (a) Prove that *localization is exact*. That is, if $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ is an exact sequence of R -modules, then $0 \rightarrow S^{-1}A \xrightarrow{S^{-1}f} S^{-1}B \xrightarrow{S^{-1}g} S^{-1}C \rightarrow 0$ is an exact sequence of $S^{-1}R$ -modules.
- (b) If B is an R -module, and A is a submodule, prove that $S^{-1}(B/A) \cong S^{-1}B/S^{-1}A$. That is, you can localize or quotient in any order.

(c) Compute the isomorphism types of the following:

$$\left(\frac{\mathbb{C}[x, y]}{(xy)} \right)_x \quad \text{and} \quad \left(\frac{\mathbb{C}[x, y]}{(xy)} \right)_{x+y}.$$

(Hint for the second one: show that $(x)_{x+y} + (y)_{x+y} = (1)$ and use the Chinese Remainder Theorem.)

Exercise 5.

- (a) Prove that a $\mathbb{C}[x]$ -module is the same thing as a $\mathbb{C}$ -vector space V and a linear transformation $X: V \rightarrow V$.
- (b) Suppose $V = \mathbb{C}^n$, and $X: V \rightarrow V$ is a linear transformation. Then (V, X) is a finitely-generated $\mathbb{C}[x]$ -module, so it is isomorphic to $\bigoplus_i \mathbb{C}[x]/(f_i(x))$ for some polynomials $f_i(x)$. In linear algebra terms, what is the relation between the polynomials f_i and the linear transformation X ?

Fugue on the G string

Exercise 6. (January 2022 Problem 3) Let R be a commutative ring.

- (a) Prove that a module M over R is projective if and only if it is a direct summand of a free module.
- (b) Let M be a projective module over R and $S \subset R$ a multiplicative set. Prove that $S^{-1}M$ is a projective $R[S^{-1}]$ -module.

Exercise 7. (January 2021 Problem 3) Two questions about projective modules.

- (a) Show that a projective module over an integral domain is torsion free.
- (b) Let $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ be a short exact sequence of finitely generated modules over a commutative, Noetherian ring. Prove the following OR give a counterexample:
 - (i) If M_1 and M_2 are projective, then so is M_3 .
 - (ii) If M_2 and M_3 are projective, then so is M_1 .

Exercise 8. (January 2025 Problem 5, parts (a), (c), (d)) Let A be a commutative Noetherian ring, and M an A -module. We say that M has finite length if there exists a positive integer n such that every chain

$$M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_k$$

satisfies $k \leq n$. The smallest such n is then called the length of M .

- (a) Prove that if M has finite length, then so does every localization $M_{\mathfrak{p}}$ at a prime ideal $\mathfrak{p}$ of A .
- (c) Prove that the following converse of (a) holds if A has finitely many maximal ideals: under this assumption, M has finite length if $M_{\mathfrak{m}}$ has finite length for every maximal ideal $\mathfrak{m}$ of A . (You can use without proof the fact that an A -module M is zero if and only if $M_{\mathfrak{m}}$ is zero for every maximal ideal $\mathfrak{m}$ of A .)

- (d) Prove that the assumption in part (c) on the finiteness of the number of maximal ideals is necessary by providing an example of a ring A and an A -module M which is not of finite length, even though $M_{\mathfrak{m}}$ is of finite length for every maximal ideal $\mathfrak{m}$ of A . (Hint: you can find such an example where A is the ring of integers.)